Project Charter: SnapFridge

Field	Details
Course	ISE304E – Software Engineering (2026–2027 Fall)
Group	SnapFridge – Group 2
Customer / Instructor	Res. Asst. Dr. Abdullah Cihan AK
Repository	https://github.com/iemiral-spec/SnapFridge
Document version / date	2.0 – 2026-11-03

1. Project Description

SnapFridge is a web application that helps users decide what to cook with the ingredients they already have. The user uploads a photo of their fridge or kitchen counter, an AI vision service lists the ingredients it recognizes, the user corrects the list, and the system suggests recipes from a recipe database. The domain is food and lifestyle; the main users are students, employees and home cooks.

2. Objectives and Outcomes

Objectives

- Turn a fridge photo into an editable ingredient list in under 30 seconds.
- Suggest recipes ranked by how many of their ingredients the user already has, and show the missing ones.
- Deliver a cloud-hosted, tested web application that meets the course's technical requirements.

Outcomes (in scope)

- Registration, login and role-based authorization (User / Admin)
- Photo upload and AI-based ingredient recognition, with user correction
- Recipe dataset (TheMealDB; Kaggle as fallback) imported into PostgreSQL
- Ingredient-based recipe suggestions with recipe details and missing ingredients
- Favorite recipes
- Admin recipe management (add, edit, delete)

Optional (only if core work is finished): analysis history, recipe categories, keyword search, AI-generated recipes.

Out of scope: native mobile app, grocery ordering or payment, social features, calorie/diet tracking, training our own AI model.

Technology: Python FastAPI (REST API), PostgreSQL, JWT authentication, React + Tailwind CSS, Render (cloud hosting), GitHub Actions (CI), a vision-capable LLM API. AI coding assistants (Claude Code / GitHub Copilot) may be used; their work, effort, human review effort and the errors they introduce will be reported.

3. Project Manager and Staff

Member	Student ID	Role
Biray Dağlı	150210211	Project Manager + Backend Developer
Ahmet İlker Yıldızlı	150210215	Configuration Manager + Frontend Developer
Umut Uğurlu	150230213	Test Manager + Backend Developer

Member	Student ID	Role
Talha Arpacık	150220213	Documentation Lead + AI Integration Developer
İdil Emiral	150230223	Communication Lead + Database Developer

4. Summary Milestone Schedule

Dates follow the course's 13-week project outline (Wednesday classes, no class on 18 November). Report deadlines are 23:59 on the date shown; the exact time of each submission should be checked on Ninova.

Week	Date (2026)	Deliverable	Customer meeting / peer scoring
3	13 Oct	Project Charter	–
4	21 Oct	–	Meeting 1 on Project Plan
5	28 Oct	Project Plan (submitted 3 Nov)	–
6	4 Nov	–	Meeting 2 on Requirements
7	10 Nov	Requirements Specification Report	Peer Scoring 1 (11 Nov)
8	24 Nov	Requirements Analysis Report	Midterm (25 Nov)
9	2 Dec	–	Peer Scoring 2
10	8 Dec	Software Design Report	Meeting 3 on Design (9 Dec)
12	22 Dec	Software Test Report, Code Freeze	Meeting 4 on Tests (23 Dec)
–	27 Dec	Codebase	–
13	30 Dec, 12:30	Final Demo and Presentation	Peer Scoring 3

5. Preliminary Cost Estimation

- **Effort:** about 170 person-hours of development work (5 members × ~10 hours/week over 9 weeks for project activities), plus time for reports and meetings.
- **Money:** no budget. We use free tiers and student offers: GitHub (Student Pack), Render free tier, PostgreSQL on Render, TheMealDB. The vision API is pay-per-use; we will pick a provider with a free tier or student credits and limit calls during development.

6. Preliminary Risks

Risk	Impact	Response
AI recognizes ingredients wrongly	Wrong recipe suggestions	User always checks and edits the list; manual entry as fallback
Vision API cost, quota or downtime	Main flow stops	Limit calls, use test images, allow manual ingredient entry
Recipe dataset licence or coverage is not adequate	Few or no matches	Switch to a Kaggle dataset or a smaller hand-picked set
Exams and other courses slow the team	Missed deadlines	Small core scope, weekly check-in, optional features dropped first
Scope grows after customer meetings	Schedule slips	New requests go to the optional list unless the customer approves a trade-off
Free hosting limits (sleep, storage)	Demo problems	Rehearse deployment early; keep a recorded demo as backup